Build Modern Apps Using Map and Charts Components

Arcgis Instant Apps

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ESRI DEVELOPER & TECHNOLOGY SUMMIT

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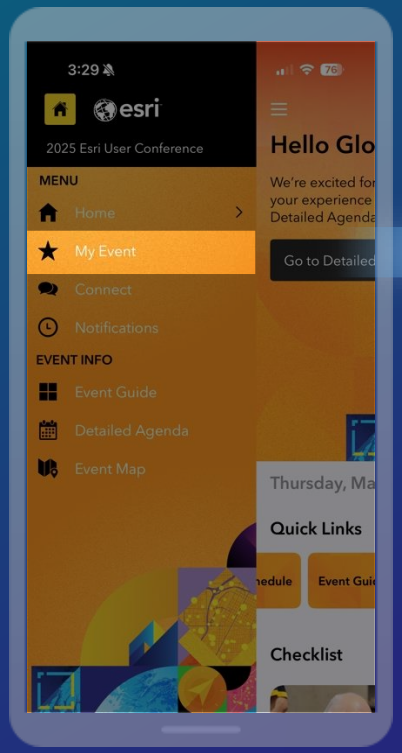

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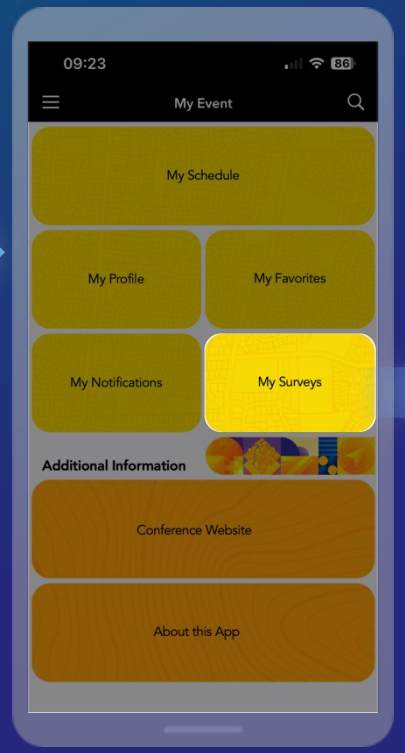

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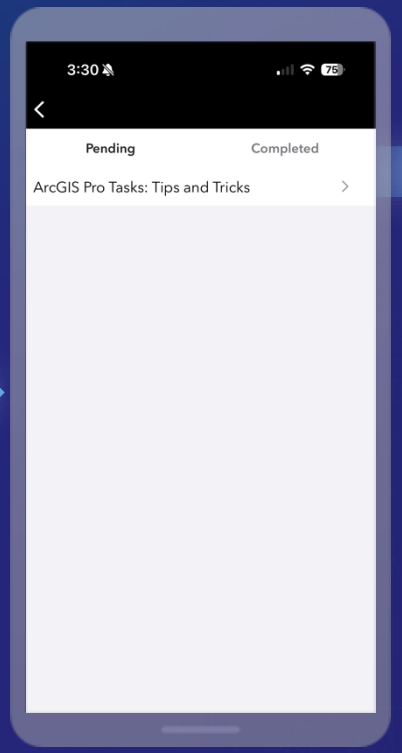
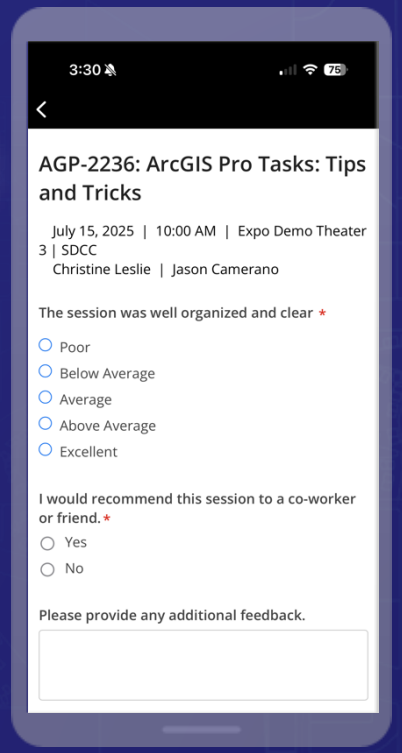


Complete the survey

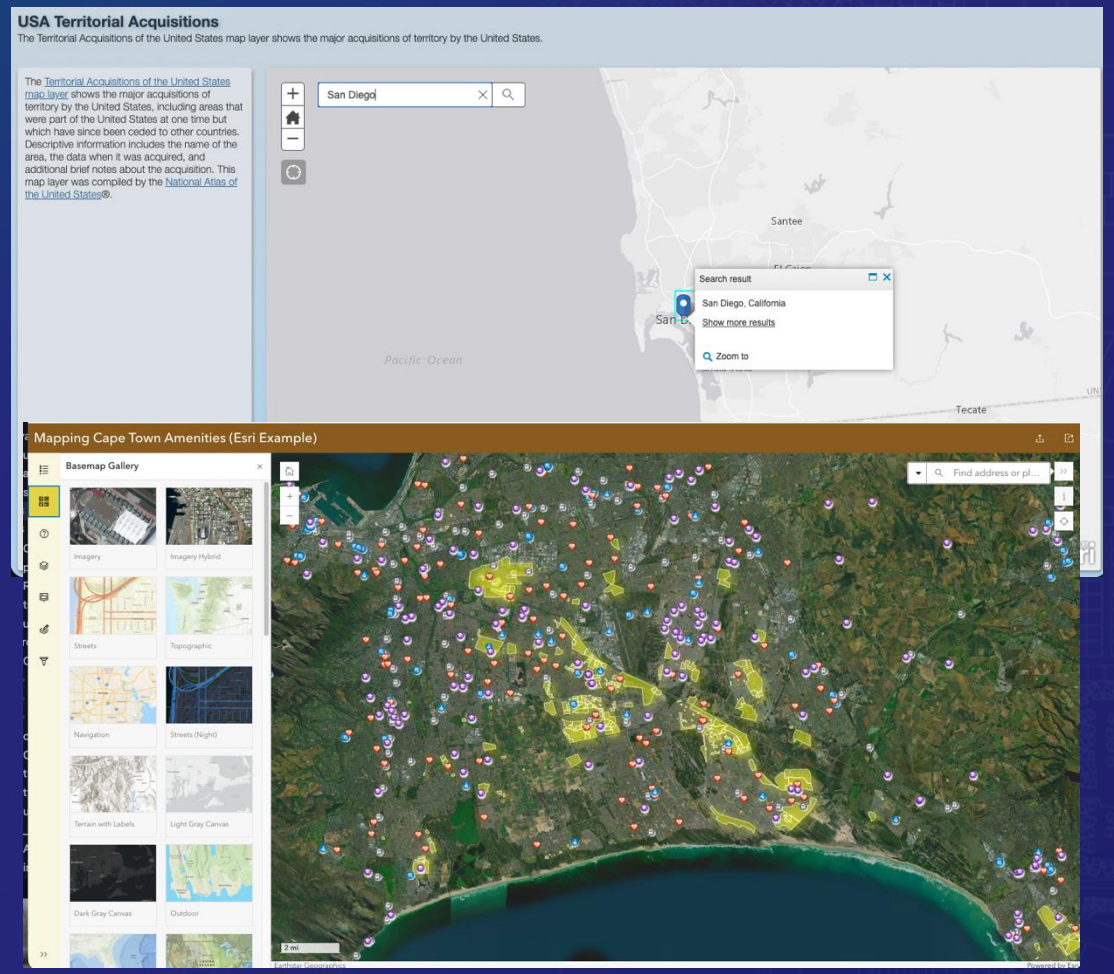


Agenda

- What are Instant Apps
- Modern App Stack
- Designing with Calcite Design System
- Building Apps with Map and Charts (beta) components
- Extending with Custom Web Components
- Theming & Branding Best Practices
- Wrap-up & Resources
- Q&A

Why this session?

- 15+ years building apps with Esri tech
- Along the way we've upgraded, refactored and migrated codebases just like you
  - 3.x to 4.x
  - Widgets to components
- Modern component driven stack: Calcite, ArcGIS Online, Maps SDK for JavaScript
- We want to share what works, what's new and what is easier now

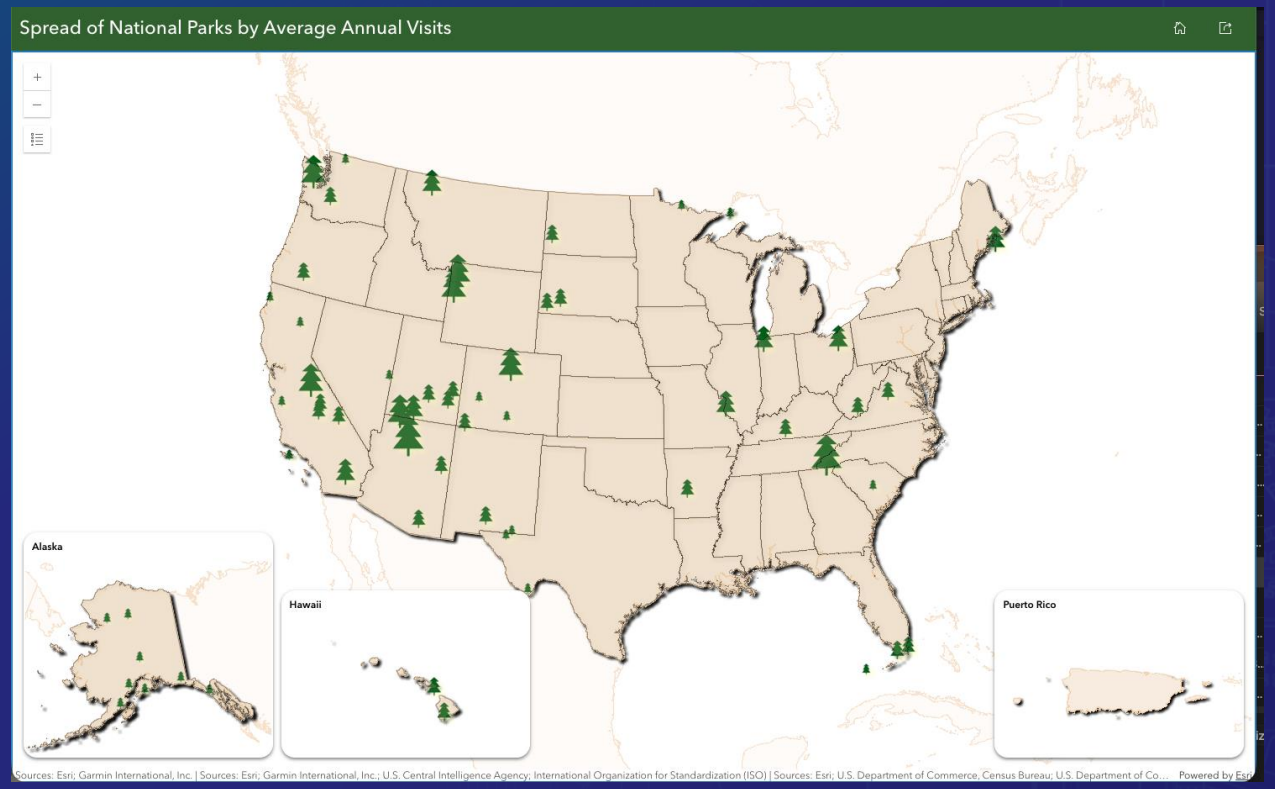


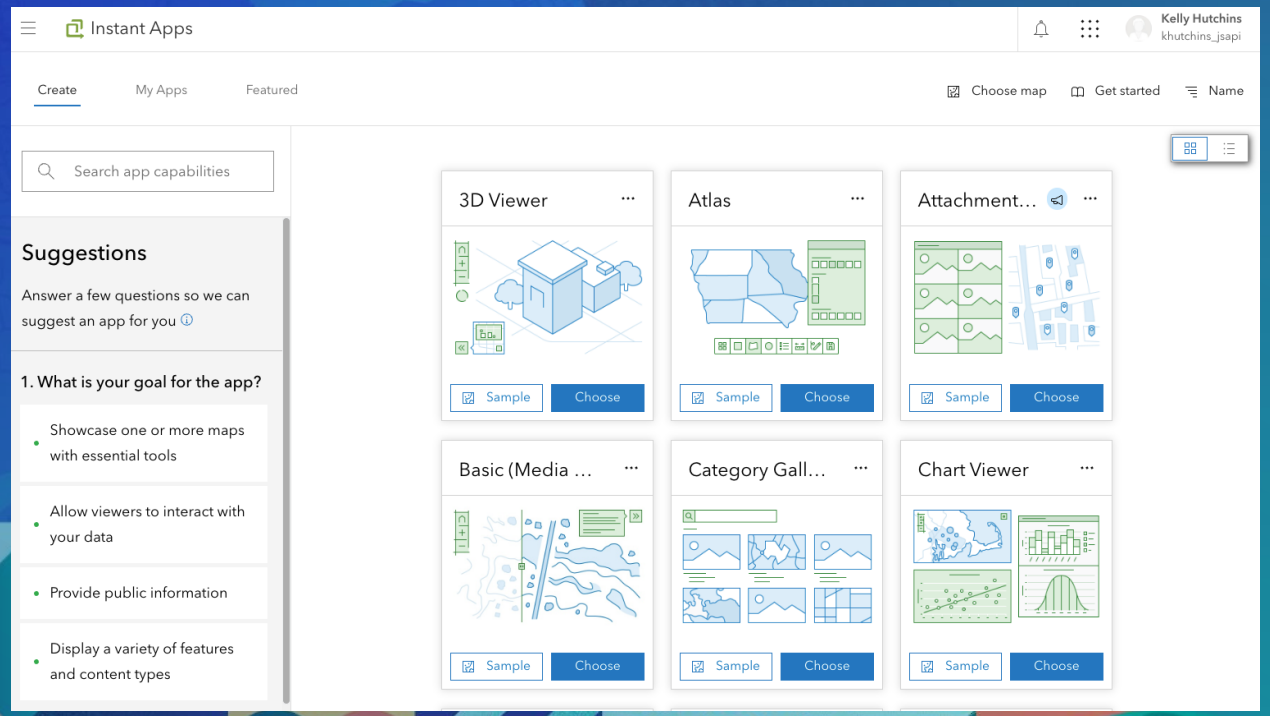
What are Instant Apps

Kelly Hutchins

What are Instant Apps?

- Share maps and apps easily
- Purpose-built app templates
- View and explore chart data
- Compare multiple maps
- Edit data and attachments
- Use inset maps for added context





Create an app

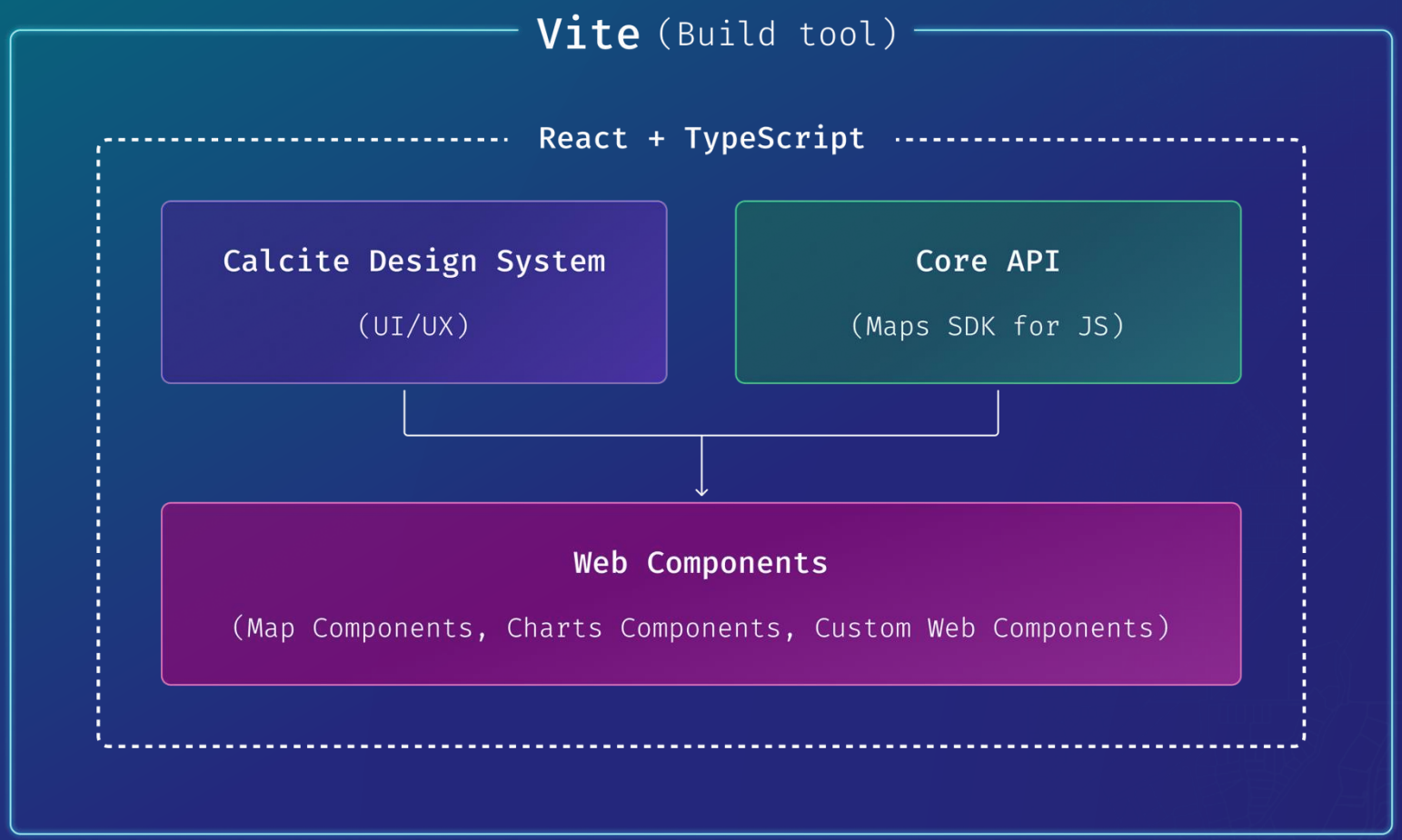
Technology Stack

- Vite – fast dev tooling
- React – component UI
- TypeScript – type safety
- Calcite Design System – consistent UX
- ArcGIS Maps SDK for JavaScript – mapping & analysis
- Custom Web Component – extend functionality

Modern App Stack

Wenping Wang

Modern App Stack Behind Instant Apps



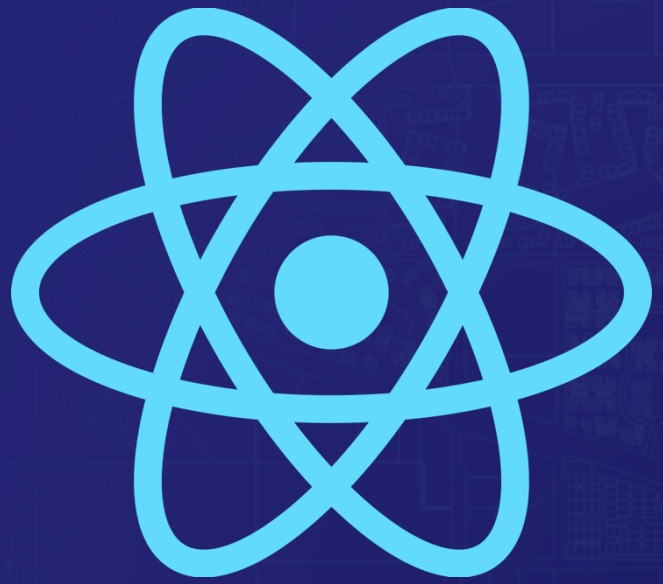
- Fast
- Modular
- Type-safe

Fast App Iteration with Vite

- Lean front-end tooling for rapid development
- Instant dev server start
- Speedy Web Compiler (SWC) integration
- Optimized production build
- CLI scaffolding for starter apps

Build Web & Native UIs with React

- Component-based UI
- Unidirectional data flow (one-way data binding)
- Virtual DOM for efficient updates
- JSX – write JavaScript in HTML-like syntax
- Popular framework with extensive community support



TypeScript for Reliability

TypeScript for Reliability

- Add types to your JavaScript code
- Catch bugs earlier
- More predictable and readable code

Key Features

- Static types
- Interfaces
- Enums

Demo: Build a Web Mapping App with @arcgis/create

Wenping Wang

Calcite Design System

Wenping Wang

Calcite Design System Overview

- Build beautiful, consistent apps easily
- Ensure consistent UX
- Accelerate development timelines
- Support accessibility & web standards
- Integrate with your framework of choice

Don’t reinvent the wheel — focus on what matters in your app, like business logic.

Demo:
Calcite Layouts

Wenping Wang

Map and Chart Components

Kelly Hutchins

ArcGIS Maps SDK for JavaScript

- Map Components
  - Drop map or scene component into app
  - Use slots to add widgets to the UI
    - Zoom, Home, Compass, Search, Legend etc.
  - Add charts components for quick data visualization

```
<arcgis-map item-id="05e015c5f0314db9a487a9b46cb37eca">
    <arcgis-zoom slot="top-left"></arcgis-zoom>
    <arcgis-legend slot="bottom-right"></arcgis-legend>
</arcgis-map>
```

Custom Web Component

- Great when logic is shared across several apps or teams
- Works regardless of framework (React, Vue, Svelte etc.)
- Use standard-based tools like Lit or Stencil to build custom components
- Package and publish internally for organization wide reuse

Framework specific logic

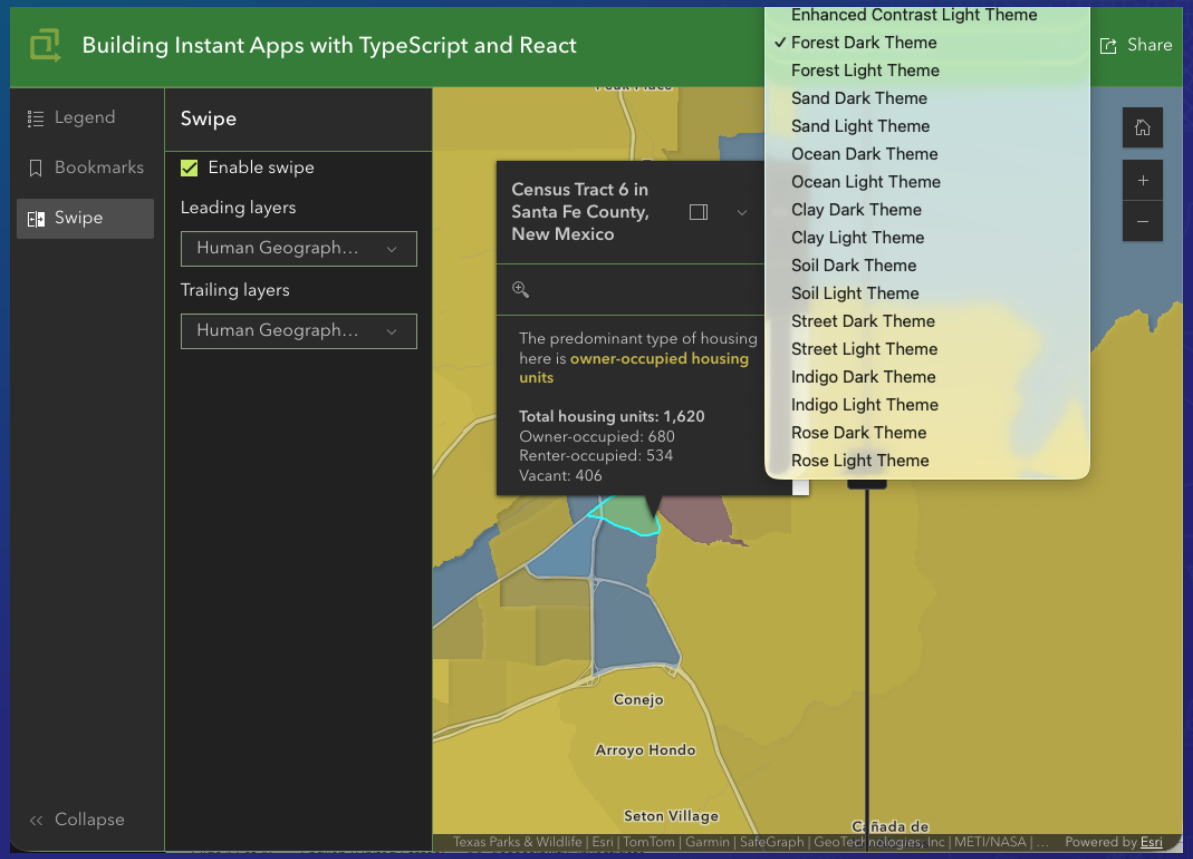
- App is tightly coupled to a specific stack (React, Vue, Angular)
- Easier integration with apps state management, routing and UI patterns
- Reusable inside other apps that use the same framework

Theming and Branding

Kelly Hutchins

Theme App

- Calcite Design Tokens
- Light/dark theme
- Calcite icons



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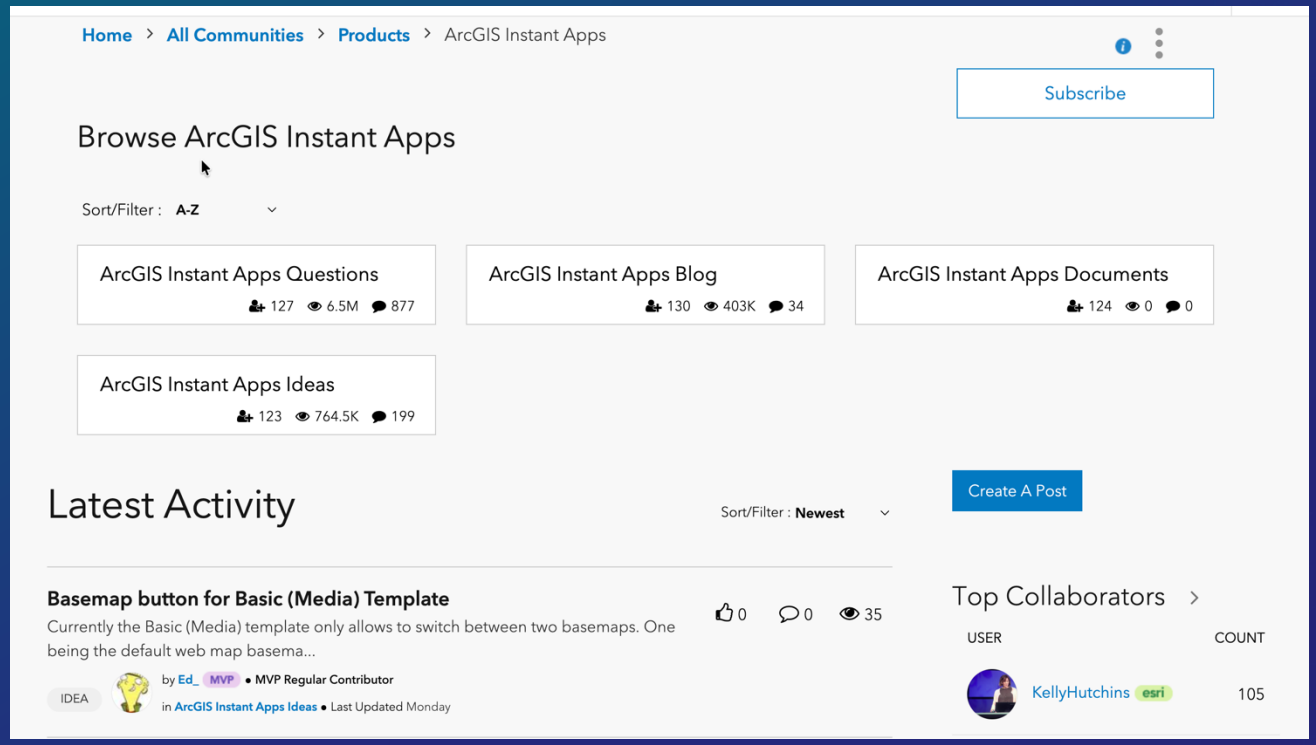
- https://community.esri.com/t5/arcgis-instant-apps/ct-p/arcgis-instant-apps



Resources

- https://www.npmjs.com/package/@arcgis/create
- https://www.esri.com/arcgis-blog/products/js-api-arcgis/developers/quick-start-new-javascript-maps-sdk-projects-with-arcgis-create
- https://developers.arcgis.com/calcite-design-system/sample-code/
- https://github.com/EsriDevEvents/dev-tech-summit-demo-templates/tree/main#